

since the massive amount of foam coming looks like toothpaste an elephant would require.

54 Mystery mystery!

Do not try to figure out what this activity does, just do it on faith. Trust us, you won't be burning down the house or blowing up your city, or doing any damage. It will be slightly messy, and get a lot of newspapers and lay them down. Get a soda bottle that's cut in half, or a soda can to hold the mixture we'll be making. If you're under 13, make sure an adult is doing all the cutting. You'll need water, vinegar, red or orange food colouring (only for show for dramatic purposes), liquid dish soap (optional) and baking soda.

Mix half a cup of water with a quarter cap each of vinegar and liquid dish soap. Add the food colour-



ing until the colour is dark enough. Next, there are two approaches here. You either add the mixture to the baking soda already poured in the container, or drop baking soda into the mixture. For a better experience we would suggest you to drop the baking soda after you've poured the liquid into the container. This is because there's very little time to react after the two mix, and since pouring the liquid will take longer, it's best to drop the baking soda using a funnel or tissue paper. As soon as you drop it, run for cover and watch the magic unfold from a safe distance of at least a few feet away. It won't kill you if you don't move, but it can mess up your clothes!

CELEBRATE A GEEKY DIWALI

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Make firecrackers

Since this magazine is going to be released in the same month as Diwali, we obviously had to have one festive fun thing here.



Of course, since there are readers as young as 8-years old that read this magazine, we're certainly not going to be telling you how to make explosives. Besides, the chemicals would be a lot harder to come by.

Instead, we're going to show you how to make really tiny firecrackers that should be safe in a normal outdoor setting. Only do this if you are old enough to, or else ask your parents to help you do this.

You can make firecrackers, from regular matchsticks. Get three or four matchboxes, take out the matchsticks and start crushing off the ends that burn – the phosphorous covering to separate it from the sticks. Also add some of the phosphorous from the sides of the matchbox by scraping it off with a sharp knife. Crush everything into as fine a powder as you can.

Now, to make the body of the firecracker, you can use small cardboard tubes, or roll up paper into tubes. Seal off one end of the tube with tissue paper and glue it all up. Next, pour the powder in the tube, leaving some space for the fuse. To make the fuse, spread the powder over a piece of paper and tightly roll it up as thin as you can. The tighter it is, the better the fuse. Insert the fuse into the main tube so that it's in contact with the powder.

Tightly pack the tube with more powder and at the end pack it in with some tissue paper. Your homemade firecracker is now ready. If you have the patience to, or have friends to help, you could make several and tie them all together into a bundle to make a more spectacular explosion.

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Black snakes

Black snakes or glow worms are one of those fireworks that don't explode. And they are easy to make from household items. The ingredients you'll need include baking soda, powdered sugar, kerosene (or lighter fluid), sand, and a bowl. Mix one part of baking soda to four parts of powdered sugar. So, if you're taking 10 g of baking soda, add it to 40 g of sugar. Mix them up



well. Pour sand into a bowl and add the lighter fluid or kerosene. The liquid is the fuel to ensure that the mixture catches fire hence, you're free to use other liquids like alcohol as well. The sand will ensure that the fire doesn't spread out. You could also just pouring the mixture outdoors on to a pile of dry mud. Drop the mixture on the sand and light it up. In just a few seconds, the mixture starts burning and a big black snake starts rising.

One thing you need to take care of is to pour the fuel on to the sand, and not on to the baking soda and sugar mixture. Make a nice wet circle of fuel in the sand, if you want a nicer snake, instead of just pouring fuel randomly.

We hope you have fun doing something different this festival season. Let us know if you have suggestions of your own to add to our list, and send us pictures of your efforts. 